

4(a)(i)	frequency (modulation)	B1
4(a)(ii)	1. zero	B1
	2. frequency (of 1.2 MHz) varies by ± 50 kHz	B1
	frequency varies (by ± 50 kHz) at a rate of 8000 times per second	B1
4(b)(i)	wavelength = $(3.00 \times 10^8) / (240 \times 10^3)$	C1
	(= 1250 m)	A1
	= 1.25 km	
4(b)(ii)	bandwidth = 30 kHz	A1
4(b)(iii)	frequency = 15 kHz	A1